D3 Code Coverage Report

Coverage of the final code on `master` at `5c8411d`, measured on 6 November 2026 with Vitest and V8. The unit run enforces an 80% threshold on all four metrics (NFR-RD-17), and the run fails below it.

Unit test coverage

Metric	Covered	Total	Coverage
Statements	6,174	7,042	87.67%
Branches	5,249	6,487	80.91%
Functions	1,228	1,383	88.79%
Lines	5,500	6,112	89.98%

Unit test coverage by area

Area	Files	Statements	Branches	Functions	Lines
Other (`src`)	2	0.00%	0.00%	0.00%	0.00%
Pages (`src/app`)	43	90.41%	81.48%	89.66%	90.94%
API routes (`src/app/api`)	48	91.42%	85.07%	89.26%	94.60%
Components (`src/components`)	77	84.35%	78.53%	84.18%	85.87%
Domain logic (`src/lib`)	60	93.21%	88.00%	96.63%	95.85%
AI features (`src/lib/ai`)	17	78.29%	69.68%	85.82%	82.05%

Lowest statement coverage

Files with at least 20 statements.

File	Statements	Statement count
`src/lib/ai/gemini-ranking.ts`	0.00%	42
`src/components/rides/JoinRequestControls.tsx`	23.40%	94
`src/components/profile/ProfileEditor.tsx`	53.70%	108
`src/components/chatbot/ChatbotInterface.tsx`	60.71%	112
`src/lib/ai/recommendations.ts`	64.81%	54
`src/app/api/rides/[id]/settlements/[occupantId]/route.ts`	66.66%	39
`src/lib/payment-settlement.ts`	70.73%	41
`src/components/rides/RideChat.tsx`	71.53%	130
`src/components/home/StudentHome.tsx`	73.68%	38
`src/lib/ai/chatbot-tools.ts`	75.24%	101

Integration test coverage

The integration suite runs the real route handlers against PostgreSQL, so it measures what runs with no database mocks. It has no threshold, because the unit run enforces the gate.

Metric	Covered	Total	Coverage
Statements	1,687	7,042	23.95%
Branches	1,197	6,487	18.45%
Functions	290	1,383	20.96%
Lines	1,547	6,112	25.31%

Full report

The browsable per-file HTML report from the CI run of 4 November 2026 is `Deliverables/D3/unit-coverage-report.zip`, and the Unit Test Report describes that run.